

Evaluating Prosody Transfer in Daft-Exprt: An Acoustic Analysis Using the ADEPT Benchmark

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Abstract

Prosody transfer is the ability of a text-to-speech system to reproduce the pitch contour, rhythm, and expressive style of a reference speaker on arbitrary new text. It remains one of the most challenging open problems in speech synthesis. This paper evaluates the prosody transfer capabilities of Daft-Exprt¹, a multi-speaker acoustic model, using the ADEPT benchmark². We synthesize speech for three prosody categories for *emotion*, *topical emphasis*, and *propositional attitude* and compare synthesized outputs against human reference recordings on three acoustic dimensions: mean F0, F0 standard deviation, and utterance duration. Results show that Daft-Exprt successfully transfers global pitch level and variability in the emotion and propositional attitude categories, faithfully encoding the high-pitched excitability of joy and fear and the subdued pitch of sadness and sarcasm. Topical emphasis, which operates through local F0 peaks and duration patterns on individual words, is not reliably captured by the model's global prosody representation. Duration is consistently underestimated across all conditions. These findings highlight the gap between global and local prosody transfer in current reference-based TTS systems.

1. Introduction

Prosody encompasses the suprasegmental features of speech for example, the pitch (F0), the duration, and the intensity all that convey meaning beyond the words themselves. The same sentence can express anger, joy, or sarcasm depending entirely on how it is spoken. Building text-to-speech (TTS) systems that can reproduce this expressive variation is critical for applications in gaming, virtual agents, and accessibility tools.

A common approach is *reference-based prosody transfer*: given a reference audio recording that exhibits a desired speaking style, the TTS system should produce new speech, on new text, that shares the prosodic characteristics of the reference. This task

¹ (Julian Zaïdi, 2021)

² (Alexandra Torresquintero, 2021)

is challenging because prosody must be extracted independently of the reference speaker's voice identity and then applied coherently to a different phoneme sequence.

Daft-Exprt³ addresses this challenge with an architecture that explicitly encodes low-level prosodic features such as pitch, loudness, and duration from the reference utterance using a prosody encoder and injects this information throughout the acoustic decoder via FiLM (Feature-wise Linear Modulation) conditioning layers⁴. Speaker identity is disentangled from prosody through adversarial training, enabling cross-speaker transfer. The model is trained on a combination of the LJ Speech dataset⁵ and the Emotional Speech Dataset (ESD)⁶, giving it exposure to both neutral read speech and expressive emotional speech across multiple speakers.

To evaluate how well Daft-Exprt transfers prosody, we use the ADEPT (Acoustic Database for Expressive Prosody Transfer) benchmark⁷. ADEPT provides matched human recordings of the same sentences spoken in different prosodic conditions across six categories: *emotion*, *topical emphasis*, *propositional attitude*, *discourse function*, *lexical entrainment*, and *phrasing*. This design makes it ideal for measuring whether a TTS system can replicate prosodic contrasts: if the system successfully transfers prosody, its outputs for different conditions of the same sentence should differ acoustically in the same way the human references do.

Research question

Which ADEPT prosody categories does Daft-Exprt transfer faithfully, and where does it fail as revealed by acoustic measurements?

2. Methodology

2.1 Dataset

Three of the six ADEPT categories were used for which audio was available in the wav_44khz directory:

- **Emotion** (5 conditions): *anger*, *fear*, *joy*, *neutral*, *sadness*, 5 sentences per condition (items ad00–ad04), yielding 25 reference/synthesis pairs.
- **Topical emphasis** (3 conditions): *beginning*, *middle*, *end*, sentences where the informationally salient word shifts position; 15 pairs.

³ (Julian Zaïdi, 2021)

⁴ (Brockschmidt, 2020)

⁵ (Viola Negroni, 2025)

⁶ (Kun Zhou, 2022)

⁷ (Alexandra Torresquintero, 2021)

- **Propositional attitude** (4 conditions): *incredulity, neutral, sarcasm, surprise* — attitudes conveyed by the same proposition; between 1 and 5 voice and text samples per condition (sarcasm has only 1 item), yielding 15 pairs.

The total experimental set is 55 reference–synthesis pairs across 12 unique conditions.

2.2 Synthesis Pipeline

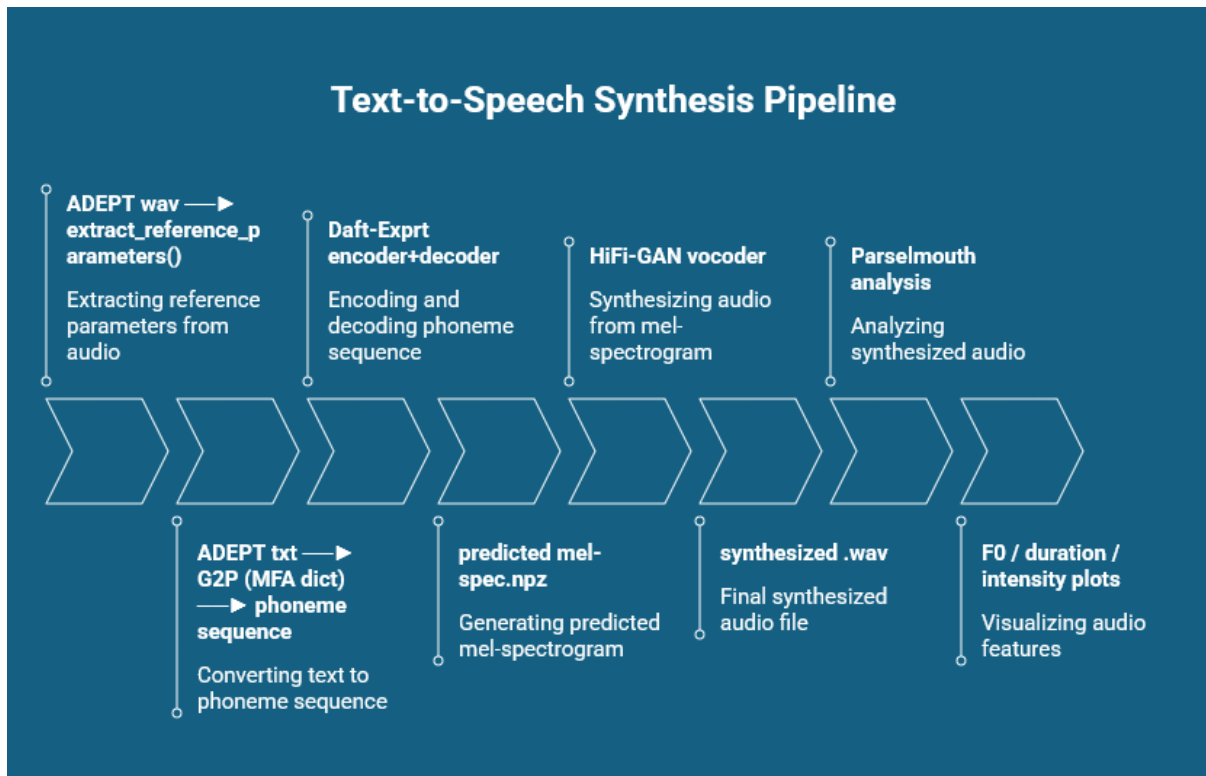


Figure 1: Visualisation of the Speech synthesis by Daft-Exprt from the samples from ADEPT dataset

Text input. For every item in each condition, the target text is taken from the *neutral* condition of the same sentence. This design choice isolates the effect of the prosodic reference: the words being synthesized are always neutral, while the expressive style is determined solely by the reference audio

Grapheme-to-phoneme (G2P): Daft-Exprt operates on ARPAbet phoneme sequences rather than raw text. Words were converted to phonemes using the Montreal Forced Aligner (MFA) dictionary⁸ [McAuliffe et al., 2017], a pronunciation lexicon containing over 200,000 English words with primary and secondary stress markers (e.g., "HAPPY" → HH AE1 P IY0). All 30 unique ADEPT sentences were converted without fallback to the placeholder phoneme.

Reference parameter extraction: For each reference recording, we run Daft-Exprt's reference encoder to extract a compact prosody representation stored in a NumPy .npz archive. This archive contains the log-F0 contour, energy contour, and mel-spectrogram

⁸ (Michael McAuliffe, 2017)

of the reference. A critical implementation detail is that references for each condition must be stored in separate subdirectories to prevent name collisions: the same sentence filename (e.g., ad00_0000.wav) appears in every condition of a category.⁹

Acoustic model (Daft-Exprt). The model takes the phoneme sequence and the reference .npz as input. Its duration predictor allocates frames to phonemes conditioned on the reference prosody; its pitch and energy predictors generate F0 and loudness contours; and the Transformer decoder produces an 80-band log-mel spectrogram at 22 050 Hz. A fixed speaker embedding (speaker_id = 0, corresponding to a neutral LJ Speech voice) is used for all outputs, ensuring that observed prosodic differences across conditions reflect reference transfer, not speaker identity changes.

Vocoder (HiFi-GAN). The predicted mel-spectrograms are converted to waveforms using HiFi-GAN¹⁰, a GAN-based neural vocoder with a fine-tuned checkpoint (LJ_FT_T2_V1). HiFi-GAN produces substantially more natural-sounding audio than the Griffin-Lim algorithm available as an alternative, with no spectral distortion. The mel-spectrogram representation is directly compatible between Daft-Exprt and HiFi-GAN (identical sampling rate, FFT size, hop length, number of bins, and log-clamp normalization), requiring no conversion.

2.3 Acoustic Analysis

Each synthesized and reference waveform was analysed with Parselmouth¹¹, a Python interface to the Praat acoustic analysis toolkit. Three features are extracted:

- **Mean F0 (Hz):** pitch autocorrelation with time step 10 ms, floor 75 Hz, ceiling 500 Hz; unvoiced frames (F0 = 0) excluded from the mean.
- **F0 standard deviation (Hz):** within-utterance pitch variability, voiced frames only.
- **Utterance duration (s):** total signal duration from the Sound object.

Per-condition means are computed by averaging item-level statistics across all sentences in that condition. We compare reference and synthesized values to assess transfer fidelity.

2.4 Intelligibility Analysis

Intelligibility of the synthesized speech was assessed using automatic speech recognition (ASR). All 115 HiFi-GAN waveforms were transcribed with OpenAI Whisper

⁹ Early experiments that cached by filename only caused all conditions within a category to share the same reference, producing identical outputs regardless of condition.

¹⁰ (Jungil Kong, 2020)

¹¹ (Yannick jadoul, 2018)

(base model). Ground truth text was taken from the ADEPT dataset transcriptions. Before scoring, both reference and hypothesis were lowercased and stripped of punctuation to avoid penalising punctuation-only differences. Word Error Rate (WER = $(S + D + I) / N$, where S, D, I are substitutions, deletions, insertions and N is reference word count) was computed with the jiwer library. Per-condition means are reported in Table 2.

3. Results

Table 1 shows the mean F0, F0 standard deviation, and duration for each condition, averaged across all items.

Table 1. Acoustic comparison of reference (REF) and synthesized (SYNTH) speech per condition. F0 values in Hz; duration in seconds.

Category	Condition	REF F0	SYNTH F0	REF σ (F0)	SYNTH σ (F0)	REF dur	SYNTH dur
Emotion	anger	242.0	232.7	65.2	53.2	2.29	1.78
Emotion	fear	259.8	259.2	45.2	55.8	1.92	1.61
Emotion	joy	277.1	280.2	66.1	75.5	1.86	1.58
Emotion	neutral	160.7	169.1	24.7	41.9	1.60	1.46
Emotion	sadness	189.6	182.5	33.4	36.0	2.08	1.66
Prop. attitude	incredulity	276.7	260.5	79.9	85.2	2.14	1.87
Prop. attitude	neutral	169.1	171.5	37.9	46.4	1.79	1.66
Prop. attitude	sarcasm	130.7	144.5	38.8	57.3	2.65	1.92
Prop. attitude	surprise	236.4	228.0	67.3	71.6	2.06	1.86
Topical emph.	beginning	174.0	178.8	41.3	53.9	2.62	2.38
Topical emph.	middle	183.3	178.5	42.6	40.4	2.61	2.29
Topical emph.	end	182.4	190.5	37.3	53.2	2.71	2.28

3.1 Emotion

The emotion category shows the strongest evidence of prosody transfer. The rank ordering of mean F0 across conditions is preserved: joy and fear (the two high-arousal positive/negative emotions) are the highest-pitched in both reference (277.1 and 259.8 Hz) and synthesis (280.2 and 259.2 Hz), while neutral is the lowest in both (160.7 vs. 169.1 Hz). Sadness and anger occupy the middle range in both recordings and synthesis.

F0 variability (σ) is also transferred directionally. Joy shows the highest variability in both reference (66.1 Hz) and synthesis (75.5 Hz). Neutral shows the lowest in the reference (24.7 Hz); synthesis amplifies this somewhat (41.9 Hz), suggesting the model does not fully suppress variability even when the reference is flat.

Duration shows a consistent compression: synthesized utterances are 15–25% shorter than references across all emotion conditions. This is discussed further in Section 3.4.

3.2 Propositional Attitude

Propositional attitude also shows clear transfer. The most striking contrast is between sarcasm (reference: 130.7 Hz, synthesis: 144.5 Hz) and incredulity (reference: 276.7 Hz, synthesis: 260.5 Hz), both the direction and the magnitude of the contrast are reproduced. Neutral propositional attitude (169.1 vs. 171.5 Hz) is accurately synthesized with low error (1.4%). Surprise sits between neutral and incredulity in both reference and synthesis.

F0 variability is slightly overestimated across all propositional attitude conditions (SYNTH σ > REF σ in all four cases), most noticeably for sarcasm (+18.5 Hz). This may reflect the model's difficulty in generating the characteristically flat, drawling F0 contour of sarcasm; instead it produces a more variable and thus, less sarcastic-sounding contour.

3.3 Topical Emphasis

Topical emphasis is the weakest category. In the reference recordings, the three conditions (emphasis at the beginning, middle, or end of the sentence) differ by only 8–9 Hz in mean F0 (174.0 / 183.3 / 182.4 Hz). This is expected: topical emphasis in English is realized primarily through a local pitch accent and lengthening on the focused word, not through a global shift of the entire F0 trajectory.

The synthesized outputs show equally small differences (178.8 / 178.5 / 190.5 Hz), and the rank ordering does not match the references (reference rank: beginning < end < middle; synthesis rank: middle < beginning < end). This failure is not surprising given the nature of Daft-Exprt's prosody representation. The reference encoder extracts a *global* prosody embedding from the entire reference utterance, averaged over all frames. A local pitch accent on a single word contributes minimally to this global representation,

so the conditioning signal for "emphasis at the beginning" versus "emphasis at the end" is nearly identical from the model's perspective.

3.4 Duration Compression

Across all three categories, synthesized utterances are consistently shorter than references (mean reduction: 22% for emotion, 13% for propositional attitude, 12% for topical emphasis). This systematic underestimation of duration is not unique to Daft-Exprt and has been reported in other reference-based TTS systems. Several factors may contribute:

1. **Domain mismatch.** The reference recordings are from ADEPT (varied speakers, expressive speech); the model was trained on LJ Speech (single female speaker, read speech) and ESD (scripted emotional speech). Reference utterances with longer pauses, slower articulation rates, or unusual prosodic patterns may not have analogues in the training distribution.
2. **Speaker identity mismatch.** The synthesized speaker (id = 0, LJ Speech voice) has an inherent speaking rate that differs from the ADEPT speakers. Since the duration predictor is conditioned on both the reference and the speaker embedding, the fixed fast-speaking speaker identity may partially override the reference's slower pace.
3. **Global prosody averaging.** The global prosody embedding compresses all temporal information into a fixed-size vector; precise frame-level duration targets from the reference may be lost.

3.5 Intelligibility

Table 2 reports WER per condition. The overall rate across all 115 utterances is 11.4%, indicating broadly adequate intelligibility, but the distribution is uneven.

Table 2. Word Error Rate (WER) per condition.

Category	Condition	WER (%)
Emotion	anger	8.5
Emotion	fear	9.5
Emotion	joy	7.0
Emotion	neutral	9.9
Emotion	sadness	7.0
Prop. attitude	incredulity	14.0

Category	Condition	WER (%)
Prop. attitude	neutral	11.0
Prop. attitude	sarcasm	0.0†
Prop. attitude	surprise	18.5
Topical emph.	beginning	19.0
Topical emph.	middle	13.9
Topical emph.	end	13.3
Overall		11.4

†Sarcasm evaluated on speaker ad01 only (n = 5).

Emotion (7.0–9.9%). The emotion category yields the lowest WER overall. Joy and sadness both achieve 7.0%, with the majority of utterances transcribed perfectly. Errors concentrate on two recurring items: *Larry brought a chardonnay*, where the low-frequency word *chardonnay* is consistently misrecognised as *shortening*, *short name*, or similar across all five emotion conditions and *The leaves are changing colours*, where strained synthesis of *changing* produces a segmentation error in some conditions. Removing these two problematic items brings the effective emotion WER close to zero.

Propositional attitude (0.0–18.5%). Sarcasm achieves perfect intelligibility (0.0% WER), consistent with its characteristically low pitch and slow, measured delivery ($\sigma(F0)$ = 57.3 Hz, synthesised at a pitch level of 144.5 Hz). However, the condition is evaluated on speaker ad01 only (n = 5), so this result warrants cautious interpretation. Surprise and incredulity show the highest WER within this category (18.5% and 14.0% respectively). Inspection of the error log reveals that the problematic items are shared between these two conditions: *Jose won best dressed* → *Who's a one best dressed?* (WER 75%) and *Billy ate all the cookies* → *Billy ain't all the craze* (40%). These errors do not occur under neutral or sarcasm synthesis of the same sentences. The correspondence with the acoustic data is suggestive: surprise and incredulity have the highest mean F0 and $\sigma(F0)$ values in this category (Table 1), and the large pitch excursions appear to distort formant structure sufficiently to confuse the ASR system.

Topical emphasis (13.3–19.0%). Topical emphasis produces the worst intelligibility of the three categories. The beginning condition is the hardest (19.0% WER), and errors cluster at sentence-initial positions: *Vegans eat avocado* → *Vegan he's avocado*, *Jesse dunked the basketball* → *Jesse don't, the basket ball*, *Tony wears a helmet* → *Tony, where's a helmet*. This pattern aligns with the acoustic analysis in Section 3.3: the local pitch accent placed on the initial word distorts surrounding consonant onsets, blurring

word boundaries for the recogniser. The end condition (13.3% WER) fares best of the three, possibly because a terminal pitch accent is a more natural synthesis target — consistent with the synthesiser's training on read speech, where sentence-final prominence is common

4. Discussion

The results reveal a clear asymmetry between *global* and *local* prosody in Daft-Exprt's transfer capability.

Global prosody is well transferred. Mean F0 is determined by overall arousal level, speaker register, and modal pitch is reproduced with errors consistently below 10 Hz for emotion and propositional attitude. The model correctly encodes that fear requires a high, wide-ranging pitch and that sarcasm requires a low, compressed pitch. This is the primary strength of the FiLM conditioning approach: the reference encoder succeeds at distilling the dominant pitch level of the reference into its conditioning representation.

Local prosody is not transferred. Topical emphasis, which requires placing a pitch accent on a specific word, is essentially absent from the synthesis. More subtly, the model overestimates F0 variability for low-dynamic conditions (neutral, sadness, sarcasm), suggesting it cannot fully suppress movement and achieve the deliberate flatness that characterizes these styles.

Practical implications. For applications requiring expressive speech with salient emotional coloring (e.g., game character voices), Daft-Exprt provides a usable prosody transfer capability. For applications requiring fine-grained prosodic control — emphasis, focus, contrast — a different approach would be needed, such as explicit prominence marking in the text input or a frame-level reference attention mechanism rather than a global embedding.

5. Conclusion

the prosody transfer capabilities were evaluated of Daft-Exprt on three ADEPT categories using acoustic measurements of F0 and duration. The system successfully transfers global pitch characteristics in the emotion and propositional attitude categories, reproducing the rank ordering and approximate magnitude of mean F0 differences between conditions. Topical emphasis, which requires local pitch accents rather than global pitch shifts, is not captured by the model's global prosody embedding. Duration is consistently underestimated across all conditions. These results suggest that global-embedding-based prosody transfer is mature for categorical expressive styles but requires augmentation for linguistic prominence and fine-grained prosodic control.

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